

5.2.2 Centripetal force

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) a constant net force perpendicular to the velocity of an object causes it to travel in a circular path	HSW1, 2, 5, 9
(b) constant speed in a circle; $v = \omega r$	
(c) centripetal acceleration; $a = \frac{v^2}{r}$; $a = \omega^2 r$	M2.4
(d) (i) centripetal force; $F = \frac{mv^2}{r}$; $F = m\omega^2 r$	
(ii) techniques and procedures used to investigate circular motion using a whirling bung.	